

Project:

Write a Pintool in Probe mode called **project.so**.

The pintool should be based on the pintool from exercise 4 that counts the number of times each basic block and each target jump was executed along with provided pintool source code "**bprofile-with-gearing.cpp**" in the moodle at: <https://moodle25.technion.ac.il/mod/resource/view.php?id=266384>

The pintool spawns a thread that, based on the gathered profiling, creates a 2nd Translation Cache called TC2, where execution continues from there.

Note that the pintool receives a knob called "**-prof_time**" that sets the number of seconds for the profiling duration (default is set to 2 seconds).

For example, the following command line:

```
<pindir>/pin -t bprofile-with-gearing.so -prof_time 4 -- ./cc1 200.i -o 200.s
```

creates the TC2 cache after 4 seconds of running the code in TC containing the profiling code that counts the execution frequency of every basic block and every target jump.

The objective of the project is to reduce the execution time of the translated code in TC2, including the time of the profiled gathering that is done at TC.

- Use the collected profiling data for each indirect jump + collect it for each indirect call as well (collected during TC execution) to apply the **de-virtualization** optimization which replaces the most frequent jump targets by direct jumps/calls.
- Apply at least one additional optimization in order to improve the TC2 execution time, from the following list:
 - Code Reordering
 - feel free to use the pintool at: <https://moodle25.technion.ac.il/mod/resource/view.php?id=136874> in order to learn how to reverse the condition in a conditional jump instruction).
 - Here is also a link to a recommended paper on Code Reordering implementation: <https://moodle25.technion.ac.il/mod/resource/view.php?id=136884>
 - Loop unrolling
 - Function inlining
 - Constant propagation
 - Or any other optimization from the optimizations taught during the classes

The pintool prints out the elapsed time (with profiling) of the translated code.

The printed elapsed time should improve performance by **more than 5%** on one of the of the following executables that are provided with the project description:

- bzip2
- cc1
- sgcc_base.mytest-m64
- sgcc_peak.mytest-m64
- cpugcc_r_base.mytest-m64

It is allowed to disable any print outs (including the profiling) in order to reach the requested performance objective.

Programming Tips:

Consider using the following available debug service functions:

1. **dump_instr_from_xedd()**
2. **dump_instr_from_mem()**
3. **dump_entire_instr_map()**
4. **dump_tc()** for TC and TC2, using the knobs: **-dump_tc** and **-dump_tc2** correspondingly.

5. `dump_orig_code()` (invoked by the `-dump_orig_code` knob)
6. `dump_prof()`

Submission requirements:

The submission of this project is **in pairs only**.

Submit 1 compressed file called **"project.zip"** into the moodle project [link](#) containing the following files:

1. The binary of your pintool **project.so** (compiled, and tested by you that it runs and gives the result).
2. A directory called: 'src' containing all the sources of your pintool along with the make files and a **README.txt** file that includes the following:
 - a. names + id numbers
 - b. compilation command
 - c. how to run the tool.
 - d. Describe the threshold used to distinguish between frequently and rarely executed instructions.

Submission deadline: midnight Thursday October 1st, 2026.

Project defense:

Before the submission date, we will publish several dates and time slots for each pair to choose from, in order to defend your project.

The defense will be done via zoom.

During defense, **each pair must prepare an 8-minutes video clip** describing:

1. The problem that you are trying to solve.
2. The solution which includes a presentable walk thru on the code implementation.
3. Measurement results in the form of **graph/s**.

Be prepared to defend your project implementation and results by answering questions during the defense session.

It is allowed to prepare backup foils, in case you feel it is needed for the defense.

Good Luck!